VICTORY 2027

FORCE WORK POWER & ENERGY

LECTURE-05 PHYSICS

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Topics To Be Covered

1. ENERGY
2. DIFFERENCE BETWEEN ENERGY AND POWER
3. MECHANICAL ENERGY
4. QUESTIONS


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Physics Wallah

ENERGY

- **A body capable of doing work is said to possess energy. The energy possessed by a body is measured by the amount of work that the body can perform.**
- When a body does work, its energy **decreases**, while if work is done on the body, its energy **increases**.
- It means that whenever work is done, there is always a **transfer of energy**. Thus **energy** and **work** are related to each other.

RELATION BETWEEN WORK AND ENERGY



TRANSFER OF ENERGY



Whenever work is done, energy is transferred.
Thus, **energy** and **work** are related to each other.

SI : Joule

CGS : Erg

$1\,J = 10^7$ erg

$P = \frac{E}{t} \Rightarrow E = P \times t$

W h

$1\,Wh = 1\,W \times 1\,h$

$= 1 \times 3600$

$1\,Wh = 3600\,J$

Commercial Unit of energy

$E = P \times t$

KW h

$1\,KWh = 1000\,W \times 3600\,s.$

$1\,KWh = 3.6 \times 10^6\,J$

1 Unit

AC → 1.5 KW

→ 10 h.

$E = 1.5 \times 10 = 15$ KWh

$= 15$ unit

UNITS OF ENERGY

The units of energy are same as that of work.

(1) The S.I. unit of energy is joule (J) and the C.G.S. unit of energy is erg, where

$$1\text{J} = 10^7 \text{ erg.}$$

(2) The bigger units of energy are : (i) watt hour (Wh) and (ii) kilowatt hour (kWh).

(i) **Watt hour:** One watt hour (1Wh) is the energy spent (or work done) by a source of power 1W in 1h, i.e.,

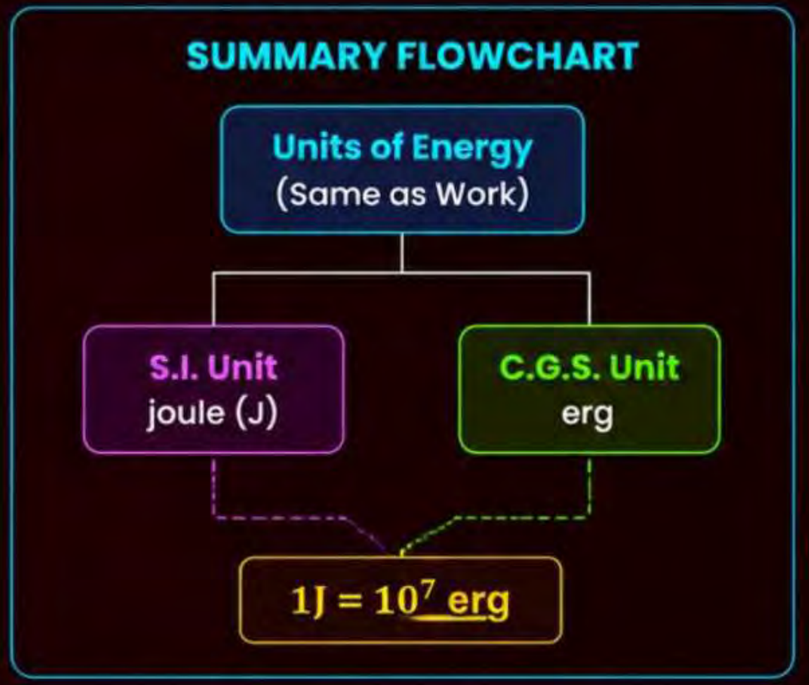
$$1 \text{ watt hour (Wh)} = 1 \text{ watt} \times 1 \text{ hour}$$
$$1 \text{ watt hour (Wh)} = 1\text{Js}^{-1} \times 3600\text{s}$$

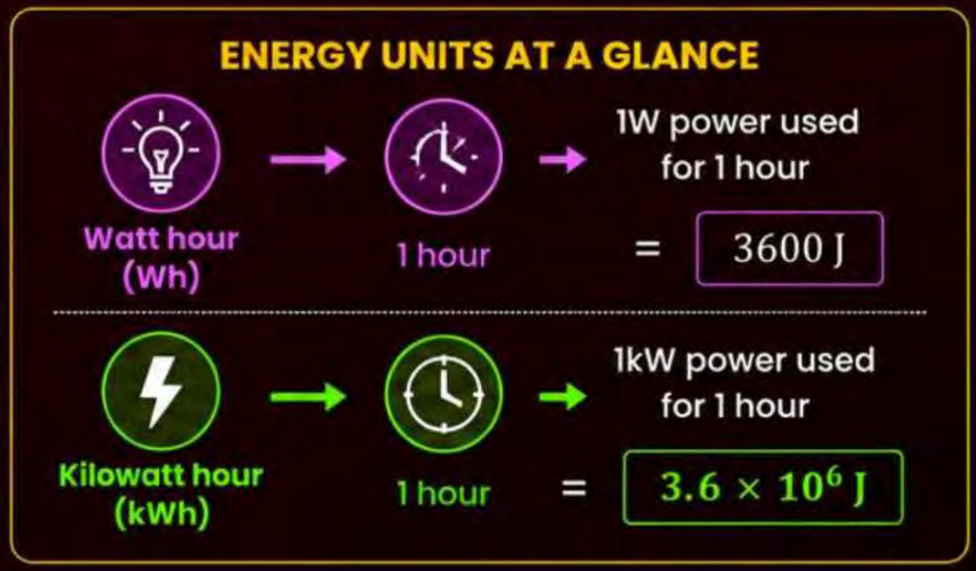
$$\textbf{1 watt hour (Wh)} = \textbf{3600J}$$

(ii) **Kilowatt hour:** One kilowatt hour (1kWh) is the energy spent (or work done) by a source of power 1kW in 1h. i.e.,

$$1 \text{ kilowatt hour (kWh)} = 1 \text{ kilowatt} \times 1 \text{ hour}$$
$$1 \text{ kilowatt hour (kWh)} = 1000\text{Js}^{-1} \times 3600\text{s}$$

$$\textbf{1 kilowatt hour (kWh)} = \textbf{3.6} \times \textbf{10}^{\textbf{6}} \textbf{ J}$$





Calorie → Heat

1g water

14.5°C ⟶ 15.5°C

$1\,Cal = 4.18\,J$

$1\,K.Cal = 1000\,Cal = 4180\,J$

Electron Volt

$1eV = 1.6 \times 10^{-19}\,J$

J
erg
Wh
KWh
Cal.
eV

UNITS OF ENERGY

(3) Calorie: Heat energy is usually measured in calorie.

1 calorie is the energy (heat) required in raising the temperature of 1g of water from 14.5°C to 15.5°C (or through 1°C).
It is related to joule as

1 calorie = 4.18J

1 kilocalorie = 1000 calorie = 4180J

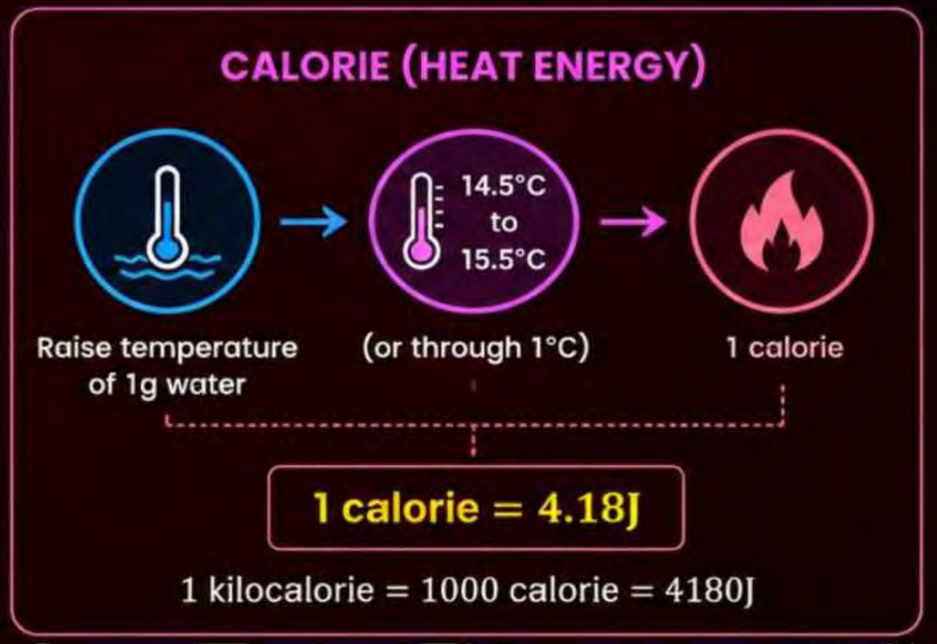
(4) Electron volt:

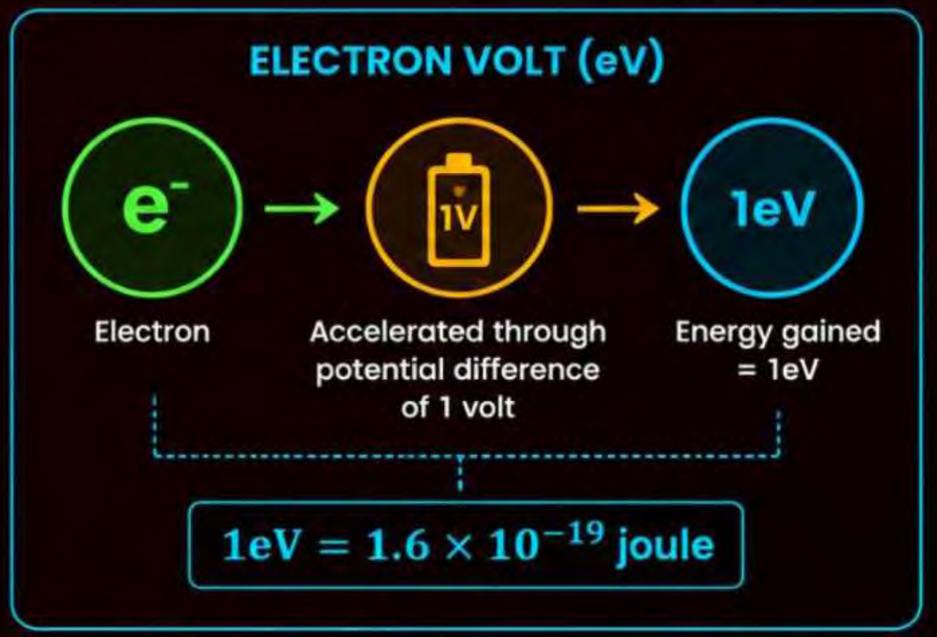
The energy of atomic particles is very small, so it is measured in electron volt **(eV)**. 1eV is the energy gained by an electron when it is accelerated through a potential difference of 1 volt. i.e.,

1eV = charge on an electron × 1 volt

$1eV = 1.6 \times 10^{-19}$ coulomb × 1 volt

$1eV = 1.6 \times 10^{-19}$ joule





DIFFERENCE BETWEEN ENERGY AND POWER

HW

	Energy		Power
1.	Energy of a body is its capacity to do	1.	Power of a source is the rate at which energy is supplied by it.
2.	Energy spent does not depend on time.	2.	Power depends on the time in which energy is spent.
3.	S.I. unit of energy is joule (J).	3.	S.I. unit of power is watt (W).

QUESTIONS

kWh is the unit of:

1. power W or kW
2. force Newton
3. energy Joule
4. none of these

QUESTIONS

What is the work done by a motor of 200W power rating in 30s?

1. 6 kJ
2. 2 kJ
3. 3.5 kJ
4. 5.5 kJ

$P = \frac{W}{t}$

$W = P \times t$

$= 200 \times 30$

$= 6000\,J$

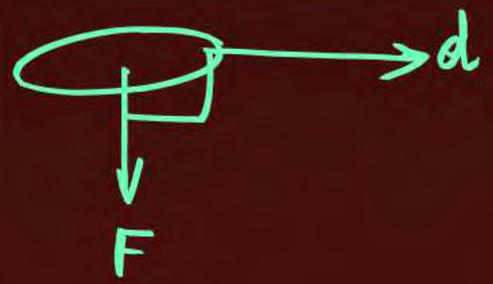
QUESTIONS

Which of the following quantities have the same unit?

1. Work and power
2. Power and energy
3. Work and energy
4. Work, power and energy

QUESTIONS

A parrot flying at a height of 300m above sea level with a force of 10N



1. does no work
2. does work equal to $\mathbf{3000}\boldsymbol{J}$
3. does negative work
4. does positive work

QUESTIONS

If a body of mass 5kg is lifted from the ground level to a height of 5m then ratio of its inertia is .....

1. 1 : 1
2. 1 : 2
3. 2 : 1
4. none of these



MECHANICAL ENERGY

The capacity to do mechanical work is called mechanical energy. The energy possessed by a body due to its state of rest or of motion, is called the mechanical energy.

Mechanical energy can be of two types :

(1) Kinetic energy → Due to Motion

(2) Potential energy → Due to position

The total mechanical energy of a body is equal to the sum of its potential energy and kinetic energy.

(1) KINETIC ENERGY



Energy possessed by a body due to its state of **motion.**

(2) POTENTIAL ENERGY



Energy possessed by a body due to its state of **rest** or its **position.**

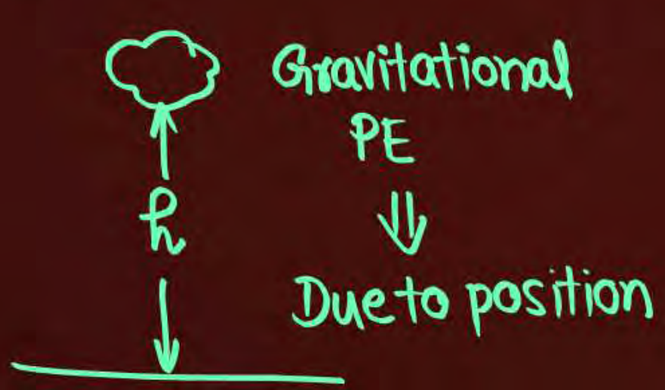
TOTAL MECHANICAL ENERGY = POTENTIAL ENERGY + KINETIC ENERGY

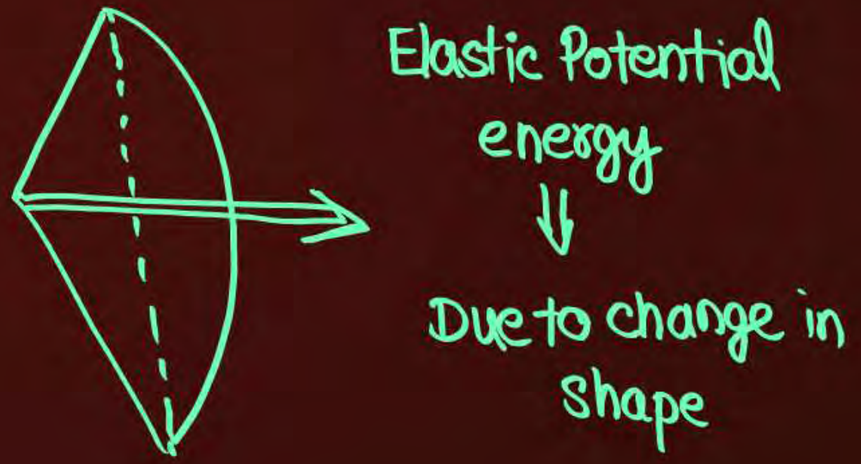
POTENTIAL ENERGY (U)

The energy possessed by a body by virtue of its changed position (or configuration) is called the potential energy.

It is usually denoted by the symbol U.

Examples: A body placed at a height above the ground, a compressed spring, a bent bow, a stretched rubber string, etc. have the potential energy.





FORMS OF POTENTIAL ENERGY (U)

In mechanics, potential energy is mainly of two kinds:

(1) gravitational potential energy due to its changed position, and

(2) elastic potential energy due to its changed configuration.



$$U_2 = -\frac{GMem}{Re+h}$$

$$U_1 = -\frac{GMem}{Re}$$

↳ Gravitational Potential energy is always -ve. It represents attraction.

Change in GPE,

$$\Delta U = U_f - U_i$$

$$= \left(-\frac{GMem}{Re+h}\right) - \left(-\frac{GMem}{Re}\right)$$

$$= \frac{-GMem}{Re+h} + \frac{GMem}{Re}$$

Extra Gyaan



Newton's Law

$$F = G\frac{m_1 m_2}{r^2}$$

Universal Gravitational Constant

$G = 6.67 \times 10^{-11}\ Nm^2/kg^2$

$$\Delta U = GMem\left(\frac{1}{Re} - \frac{1}{Re+h}\right)$$

$$= GMem\left(\frac{(Re+h) - Re}{Re(Re+h)}\right)$$

$$= GMem\ \frac{h}{Re(Re+h)}$$

$$= m\left(\frac{GMe}{Re(Re+h)}\right)h$$

$$\boxed{\Delta U = mgh}$$



$\Delta U = mgh$

$= mg(5)$

$= 2 \times 10 \times 5$

$= 100$ J.

$\boxed{mgh}$ PE ✗

Change in PE ✓

FORMS OF POTENTIAL ENERGY (U)

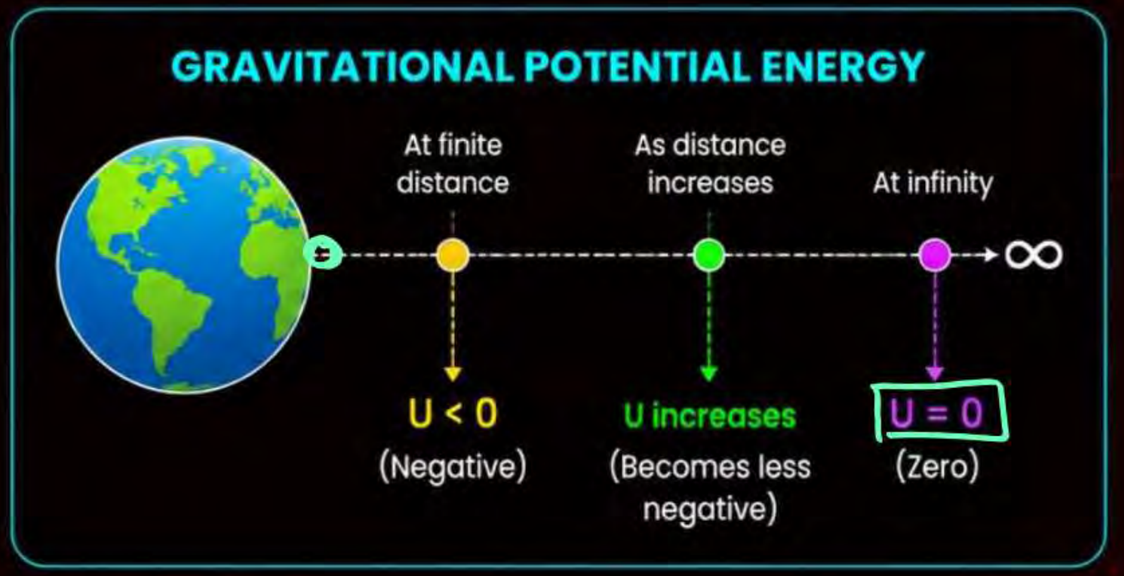
(1) Gravitational potential energy:

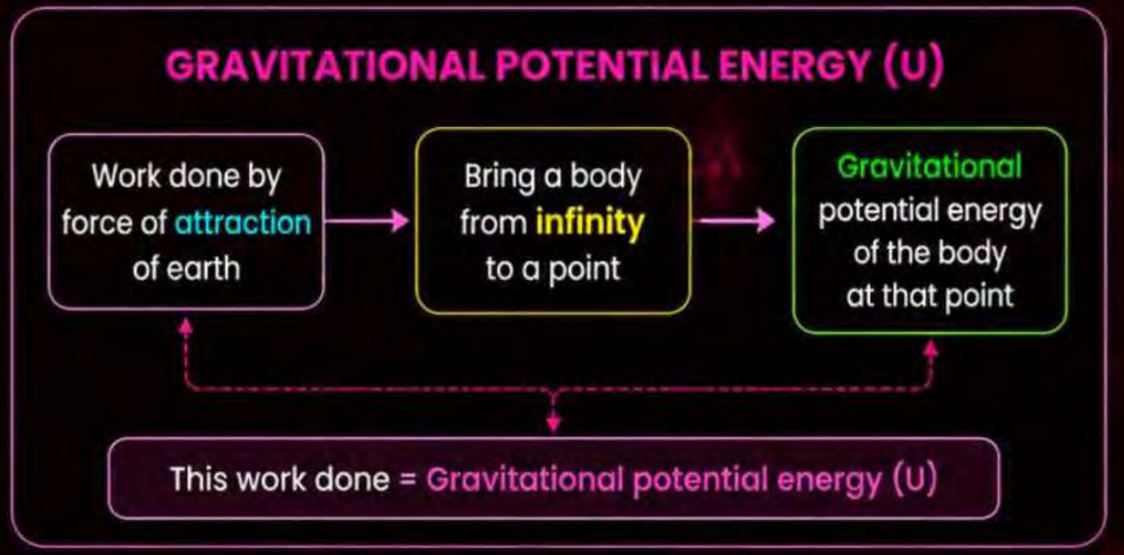
The potential energy possessed by a body due to force of attraction of earth on it, is called its gravitational potential energy.

- The gravitational potential energy of a body is **zero** when it is at infinity because the force of attraction of earth on the body is then zero.
- At a **finite** distance from earth, the gravitational potential energy of the body is **negative.**

As the distance of body from the surface of earth increases, the force of attraction of earth decreases and its gravitational potential energy increases (i.e. the negative value of gravitational potential energy decreases).

The gravitational potential energy of a body at a point is the amount of work done by the force of attraction of earth in bringing that body from infinity to that point.





FORMS OF POTENTIAL ENERGY (U)

(2) Elastic potential energy:

When an external force is applied on a body, its configuration (or shape) changes. **Elasticity** is the property by virtue of which a body regains its original configuration on removal of the external force.

The potential energy possessed by a body in the deformed state due to change in its configuration is called the **elastic potential energy**. It is equal to the amount of work done in deforming the body (or in changing the configuration of the body).

ELASTIC POTENTIAL ENERGY



Elastic potential energy is the energy possessed by a body in the **deformed state** due to change in its configuration and is equal to the amount of **work done in deforming** the body.

GRAVITATIONAL POTENTIAL ENERGY AT HEIGHT

The gravitational potential energy of a body at a height above the ground is measured by the amount of work done in lifting it up from the ground to that height against the force of gravity (assuming that its gravitational potential energy on the ground is zero).

This work is done in lifting the body up and it gets stored in the body in the form of its gravitational potential energy when it is at height h.

QUESTIONS

Gravitational potential energy = _____

1. mgh
2. $2\,mgh$
3. $1/2\,mv^2$
4. $1/2\,mgh$

QUESTIONS

Energy is _____

1. the capacity to use power.
2. capacity to produce power
3. capacity to do work
4. capacity to do work per unit time.

QUESTIONS

The two basic forms of mechanical energy are:

1. PE and KE
2. Sound and Light
3. Heat and tidal
4. Solar and wind

KINETIC ENERGY

The energy possessed by a body by virtue of its state of motion is called the kinetic energy. It is usually denoted by the symbol **K**.

Examples: A fast moving stone has the capacity of breaking a window pane on striking it and thus it has the kinetic energy. Similarly, a car in motion, moving hands of a clock, the bullet fired from a gun, a rolling ball, an apple falling from a height, etc. have the kinetic energy.



A fast moving stone



A car in motion



Moving hands of a clock



The bullet fired from a gun



A rolling ball



An apple falling from a height



All have kinetic energy

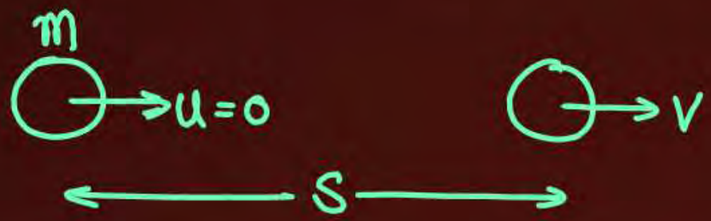


Kinetic energy is the energy possessed by a body due to its state of **motion.**

EXPRESSION FOR THE KINETIC ENERGY

The kinetic energy possessed by a moving body is equal to the amount of work which the moving body can perform before coming to rest.

Suppose a body of mass m is moving with a velocity v. It is brought to rest by applying a constant opposing force F.



$v^2 = u^2 + 2as$

$v^2 = 0 + 2as$

$$s = \frac{v^2}{2a}$$

$W = Fs$

$= (ma)\left(\frac{v^2}{2a}\right)$

$$W = \frac{1}{2}mv^2 = K.$$

RELATIONSHIP BETWEEN KINETIC ENERGY AND MOMENTUM

$$p = mv \quad \text{---} (1)$$

$$K = \frac{1}{2}mv^2 \times \frac{m}{m}$$

$$= \frac{1}{2}\frac{(mv)^2}{m}$$

$$\boxed{K = \frac{p^2}{2m}}$$

$$p^2 = 2mK$$

$$\boxed{p = \sqrt{2mK}}$$

HOMEWORK

- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.
- COMMENT YOUR RESPONSE ABOUT THIS CLASS.

QUESTIONS

1. The work done by the heart is $1\,\mathrm{J}$ per beat. Calculate the power of the heart if it beats 72 times in 1 minute.

2. Express $5\mathrm{kWh}$ into joule.

3. The energy of an electron is $4.0 \times 10^{-19}\mathrm{J}$. Express it in *e*V.

$K = \frac{1}{2}\frac{m^2v^2}{m} = \frac{p^2}{2m}$

Thank You